N83 K128 split root index: result

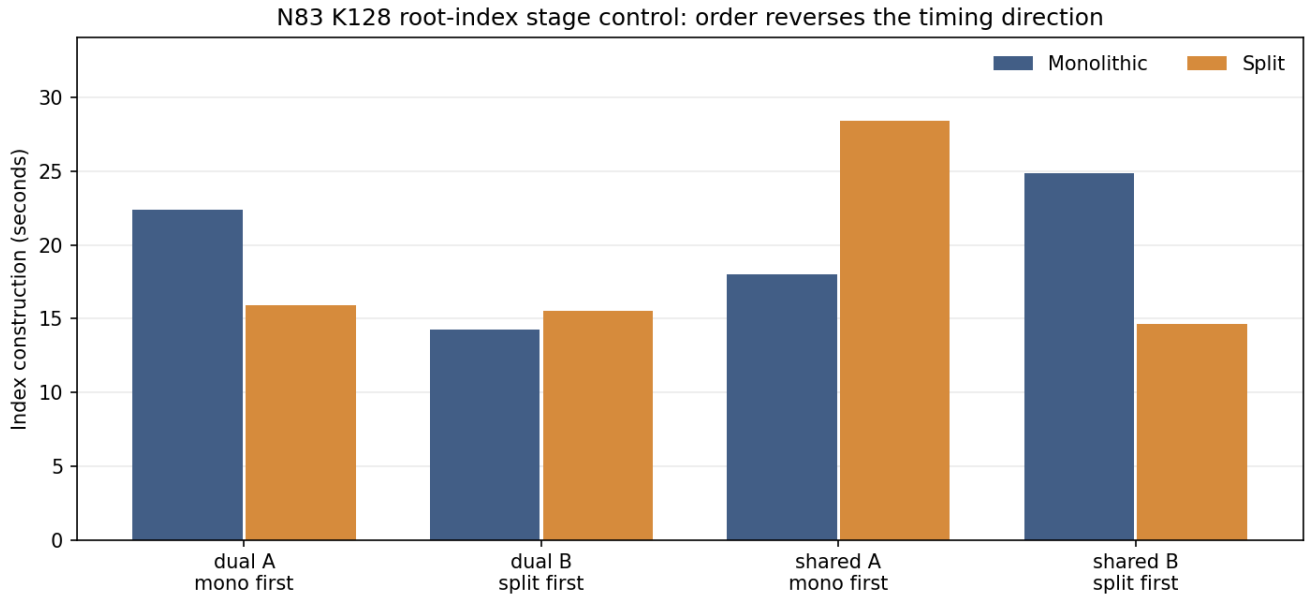
The split exact-root construction preserved the K128 base, state set, and root-entry count. Index-build timing changed direction with run order: the paired local timings do not establish a stage speedup.

At K128 the monolithic five-byte table allocated 10,485,760 bytes and the split tables allocated 13,107,200 bytes. Shared Bloom allocated 2,097,152 bytes, matching monolithic and saving 524,288 bytes compared with separate split-table filters. The frozen K4405 layout formula saves 128 MiB of Bloom space against separate filters and adds 671,088,640 table bytes against monolithic.

Status: STAGE_DIAGNOSTIC_ONLY. Candidate ID and single-target online speedup remain unset. The local full-size K4405 monolithic run reached its wall cap, and the split run stopped at its scratch-disk guard before an online row. Tier 1 paired measurement is the next gate.

The four-thread K128 replay passed exact index identity for both layouts. Its result and verification receipt are under runs/k128_root_threads4. The Tier 1 K5086 protocol freezes four index-construction threads.

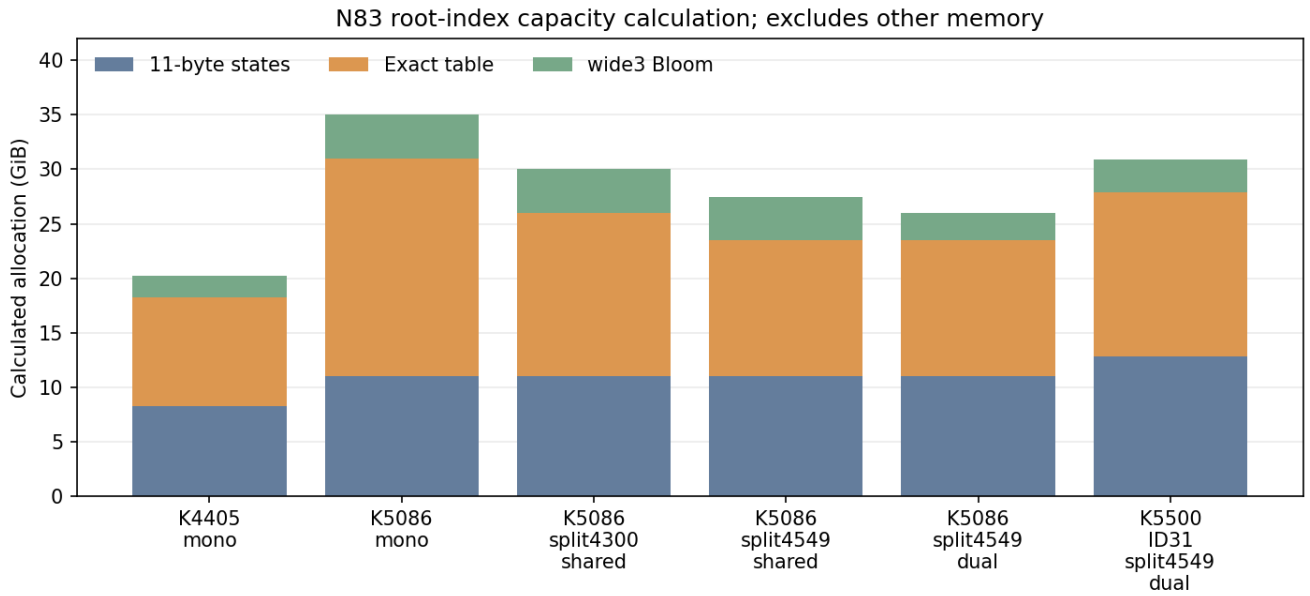
Order	Mode	Mono (s)	Split (s)	Faster stage
k128_a	dual	22.393	15.919	split
k128_b	dual	14.285	15.502	mono
k128_shared_a	shared	18.030	28.390	mono
k128_shared_b	shared	24.862	14.632	split



Source: research/n83_two_tier_root_20261021/runs/{k128_a,k128_b,k128_shared_a,k128_shared_b}/result.json; same study directory contains inputs, stderr, stdout, and runner copies.

Calculated N83 split-index capacity

The 30-bit state-ID arithmetic admits K5086 and excludes K5087 and above. With primary-first lookup, K4549 is the unique split point meeting the 2^31 primary and 2^29 extension slot capacities. Five-byte table storage is 12.5 GiB, versus 15 GiB at split K4300 and 20 GiB monolithic. Separate wide3 filters save 1.5 GiB against the union filter at K5086. The K5086 point-count formula projects 844,276 points; enumeration and subgroup checks remain to be run.



Calculated arrays only. The 11-byte states, five-byte table, and separate wide3 Blooms total 26.00 GiB at K5086/K4549; a union Bloom raises this to 27.50 GiB. Other memory and peak RSS are excluded. No K5086 timing or target result is inferred.

An opt-in 31-bit state ID extends the five-byte format. The K5500/K4549 separate-filter three-array calculation is 30.86 GiB. The 913,000-point K5500 base passed independent subgroup, orbit, and distinctness replay; full-index peak RSS, rank, and online time remain open. The pinned Lean module checks the ID31 formula bound and byte sum.

Source and proof map: research/n83_two_tier_root_20261021/CAPACITY.md and Capacity.lean; Rust source: examples/koblitz_orbit_dlp_fast.rs.